

Thomas Randall

Tucson, AZ 85719 | (520) 386-5663 | thomasrandall121@gmail.com

Profile

Meticulous ML engineer and Data Scientist with over ten years of experience in healthcare, supply chain sport and fleet management. Strong background in data preprocessing, feature engineering and model evaluation. Skilled in using frameworks like Tensorflow, PyTorch and Scikit-learn, and programming languages such as Python, R, Java, and C#. Continuously learning new techniques and staying updated with the latest advancements in Machine Learning and Data Science.

Experience

Senior Machine Learning Engineer | Curai Health | Remote October 2022 – present

- Designed and deployed NLP-based patient triage models, reducing physician response time by 30% and improving diagnostic accuracy by 20%.
- Created an extractive and abstractive summarization model using BERT & T5, reducing physician documentation time by 50% while maintaining 90%+ accuracy.
- Implemented an active learning framework that reduced data annotation costs by 40%, while improving classification accuracy from 82% to 91%.
- Engineered a Bi-LSTM + CRF-based model to segment conversations into history, symptoms, and treatment sections, increasing retrieval accuracy by 35%.
- Designed an end-to-end MLOps pipeline using AWS SageMaker and Kubernetes, reducing model deployment time from 3 weeks to 3 days.
- Developed federated learning models for decentralized training across hospitals, ensuring HIPAA compliance while maintaining 99% model performance parity with centralized training.
- Created an Explainable AI (XAI) framework using SHAP & LIME, increasing physician trust and AI adoption by 45% in diagnostic workflows.
- Implemented bias mitigation techniques (re-weighting & adversarial debiasing), reducing disparities in AI-driven diagnosis accuracy across demographic groups by 25%.
- Designed an automated ML monitoring pipeline that detects model drift in real-time, reducing incorrect AI-driven recommendations by 30%.
- Integrated AI-driven clinical decision support into physician workflows, reducing misdiagnosis rates by 18% and improving treatment recommendations.
- Published three peer-reviewed papers in top AI and healthcare conferences, enhancing Curai Health's reputation in AI-driven telemedicine.
- Spearheaded collaboration between ML, software engineering, and clinical teams, reducing AI feature development time by 40% and accelerating product release cycles.

Data Scientist | Life Protect 24/7 | Remote October 2021 – Sep 2022

- Built an ETL pipeline capable of processing millions of medical alert signals per day, reducing data processing time by 40%.
- Migrated raw healthcare data to a structured data warehouse using Apache Spark and Delta Lake, improving query performance by 60%.
- Developed automated data validation and anomaly detection rules, reducing data inconsistencies by 35% across emergency call records.
- Built a data preprocessing pipeline that removed noise and inconsistencies from IoT sensor data, improving model accuracy by 20%.

- Built a **data preprocessing pipeline** that removed noise and inconsistencies from IoT sensor data, improving model accuracy by **20%**.
- Implemented an MLOps workflow using Kubeflow and CI/CD pipelines, reducing ML model retraining time by 50% while ensuring model performance remains optimal.

Data Scientist | GPS Insight | Hybrid

Oct 2019 – Sep 2021

- Developed a real-time ETL pipeline using Kafka & MQTT, Apache Flink, AWS IoT Core to ingest real-time video, and sensor data from vehicles, processing 6PB+ of data daily.
- Implemented a data preprocessing feature to extract patterns from engine performance logs, and identify anomalies in sensor readings, achieving 90% accuracy.
- Leveraged autoencoder to cleanse missing feature and remove noise from sensor data of IoT devices, ensuring more precise conclusions.
- Implemented feature engineering techniques using time-series aggregation, rolling windows, and Fourier transforms to enhance model performance.
- Built and optimized CNN-based computer vision models to detect drowsiness, distraction, and risky driving behaviors with 90%+ accuracy using real-time dashcam footage.
- Designed LSTM and anomaly detection models to predict vehicle failures, reducing unexpected breakdowns by 30% and optimizing fleet maintenance schedules.

Machine Learning Engineer | AIScout | On-site

July 2017 – Sep 2019

- Developed a pose estimation model using Mediapipe and OpenPose, achieving 95%+ accuracy in human motion tracking, used in AI-driven athlete performance analytics.
- Optimized a computer vision pipeline with YOLOv8, improving object detection accuracy by 30% and reducing inference latency using TensorRT.
- Engineered a personalized recommendation system using collaborative filtering with a hybrid Transformer model, increasing customer engagement rates by 35%.
- Developed a cloud-based data lake using Apache Parquet and AWS Glue, reducing data retrieval times by 50% and improving analytics speed for business intelligence.

Machine Learning Engineer | Vorto.ai | On-site

Sep 2015 – June 2017

- Designed and deployed a reinforcement learning model for supply chain optimization, reducing logistics costs by 20% and improving delivery efficiency by 15% through real-time route planning.
- Developed an unsupervised anomaly detection system using Autoencoders and Isolation Forests, detecting 95% of system anomalies and reducing unplanned downtime by 30%.
- Built an AutoML pipeline using Optuna, automating hyperparameter tuning across multiple ML models, improving predictive accuracy by 15% while cutting model iteration time in half.
- Created a time-series forecasting model using Temporal Fusion Transformers (TFT), improving demand prediction accuracy by 25%, leading to better inventory planning.

Education

Master of Science (M.S.) Data Science

September 2013 – May 2015

Clarkson University | Potsdam, NY

Skills & abilities

Programming & Development

- **Languages:** Python, R, SQL, Java, C++, Scala
- **Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Scikit-Learn, XGBoost, OpenCV, Pandas, NumPy, SpyCy
- **APIs & Web Services:** RESTful APIs, GraphQL, gRPC
- **Development Tools:** Jupyter Notebook, VS Code, PyCharm

Machine Learning & Deep Learning

- Supervised & Unsupervised Learning (Regression, Classification, Clustering)
- Deep Learning (CNNs, RNNs, Transformers, Autoencoders)
- Computer Vision (Pose Estimation, Object Detection, Image Classification)
- Natural Language Processing (BERT, GPT, Named Entity Recognition, Sentiment Analysis, NLTK)
- Reinforcement Learning (Deep Q-Network (DQN), Proximal Policy Optimization (PPO))
- Time-Series Forecasting (LSTMs, Temporal Fusion Transformers, ARIMA)
- Anomaly Detection (Isolation Forest, Autoencoders, One-Class SVM)
- Recommendation Systems (Collaborative Filtering, Content-Based Filtering, Hybrid Models)

MLOps & Model Deployment

- End-to-End ML Pipelines (Kubeflow, MLflow, TensorFlow Extended (TFX))
- Model Deployment (AWS SageMaker, FastAPI, Flask, ONNX, TensorRT)
- CI/CD for ML Models (GitHub Actions, Docker, Kubernetes, Terraform)
- Model Optimization (Quantization, Pruning, Distillation, Parallelism)

Big Data & Data Engineering

- ETL Pipelines (Apache Airflow, Apache Spark, Kafka, Snowflake)
- Distributed Computing (Hadoop, Spark)
- Cloud Data Warehouses (AWS Redshift, BigQuery, Databricks)
- Data Storage & Retrieval (PostgreSQL, MongoDB, Apache Parquet)
- Real-Time Data Processing (Flink, Spark Streaming, Redis)

Cloud & DevOps

- **Cloud Platforms:** AWS (S3, EC2, Lambda, RedShift, Glue, SageMaker), GCP, Azure (Databrick, Data lake)
- **Containerization & Orchestration:** Docker, Kubernetes
- **Infrastructure as Code:** Terraform, CloudFormation
- **Monitoring & Logging:** Prometheus, Grafana, ELK Stack

Soft Skills

- Technical Leadership & Mentorship
- Cross-Functional Team Collaboration
- Problem-Solving & Analytical Thinking
- Agile & Scrum Methodologies